

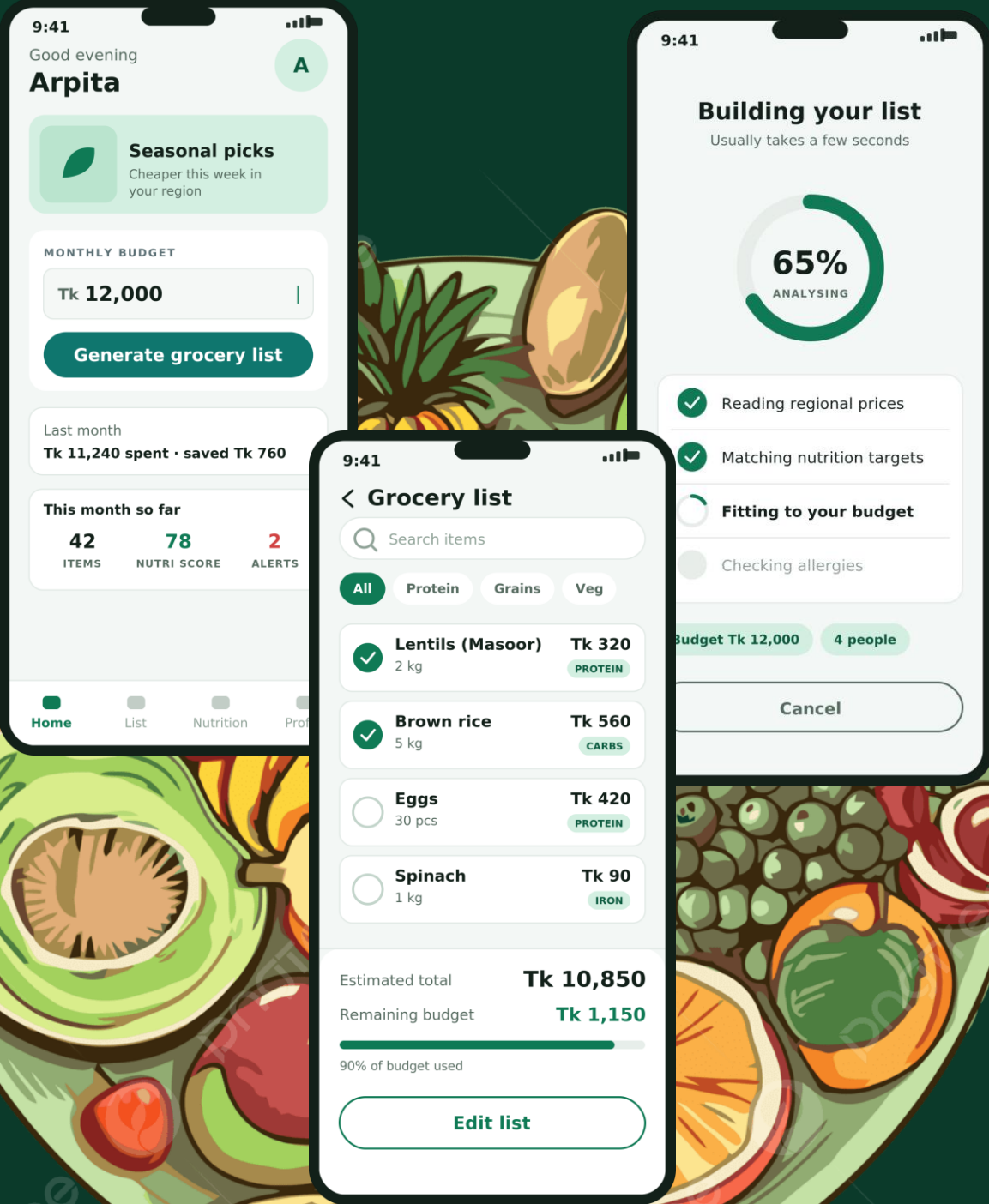
NutriCart AI

UI Flow

Twenty-one transitions across twelve screens. Each one names the element that fires it, the destination, the navigation route, the arguments carried, and what the back button does.

NAME	EXAM ROLL	ROLL	PLATFORM
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Mobile Application Development (CSE-410) · 4th Year 1st Semester
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Every flow slide has the same shape: source screen, trigger, destination screen, and the navigation contract between them.

1 Two screens, one transition

Left is where you are, right is where you land. The numbered badge marks the element that fires it.

2 Navigation is a route

Compose Navigation is given as a route string with arguments — what you type into the NavHost.

3 Not every arrow navigates

Four transitions change state without leaving the screen. Those are marked STATE CHANGE and add nothing to the back stack.

4 Back is specified

Some transitions clear the stack with popUpTo, some pop one entry, some return a result. Never assume the default.

5 Arguments are the contract

Only the listed arguments cross the boundary — usually a listId or an itemId, never a whole object.

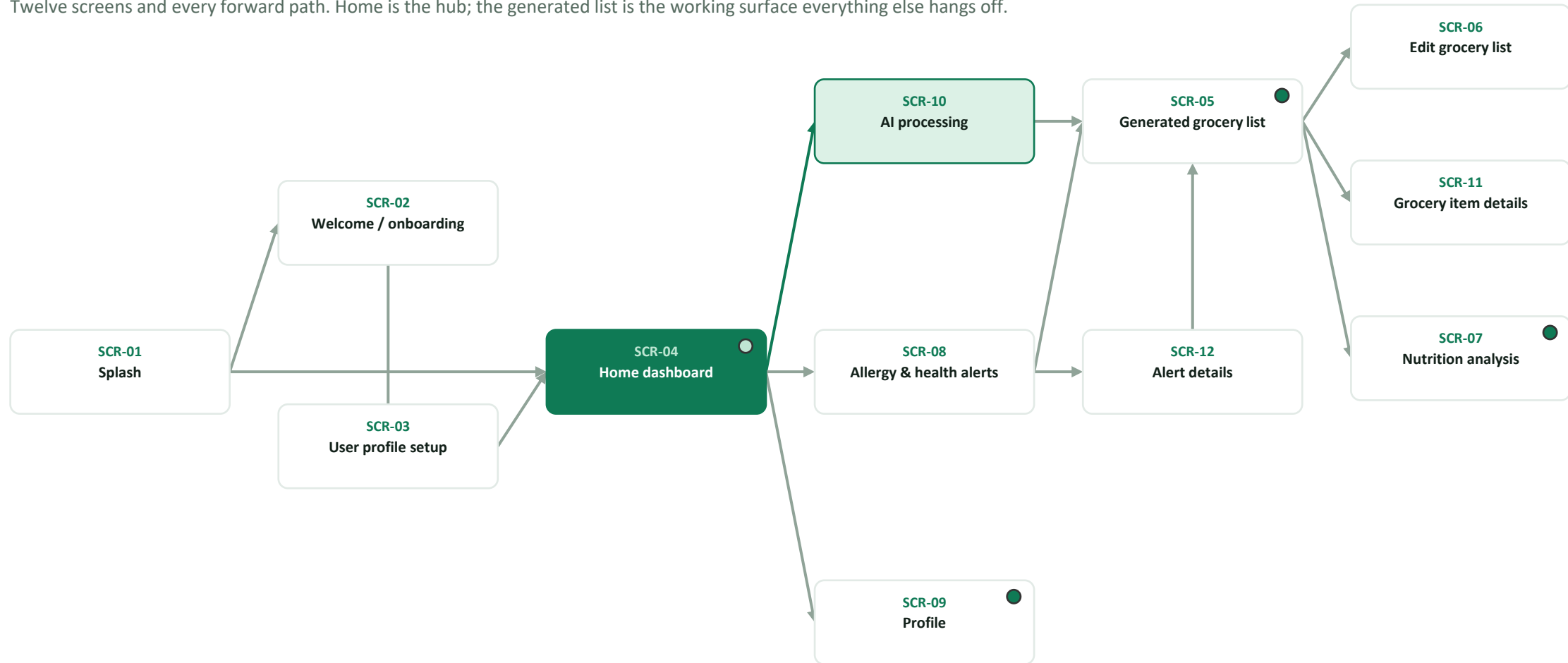
6 Three screens are new

SCR-10, SCR-11 and SCR-12 appear in this document because the flow spec requires them. They need adding to the Screen Details deck.

Solid arrow = forward navigation. Dashed = returns with a result. STATE CHANGE = same screen, new state.

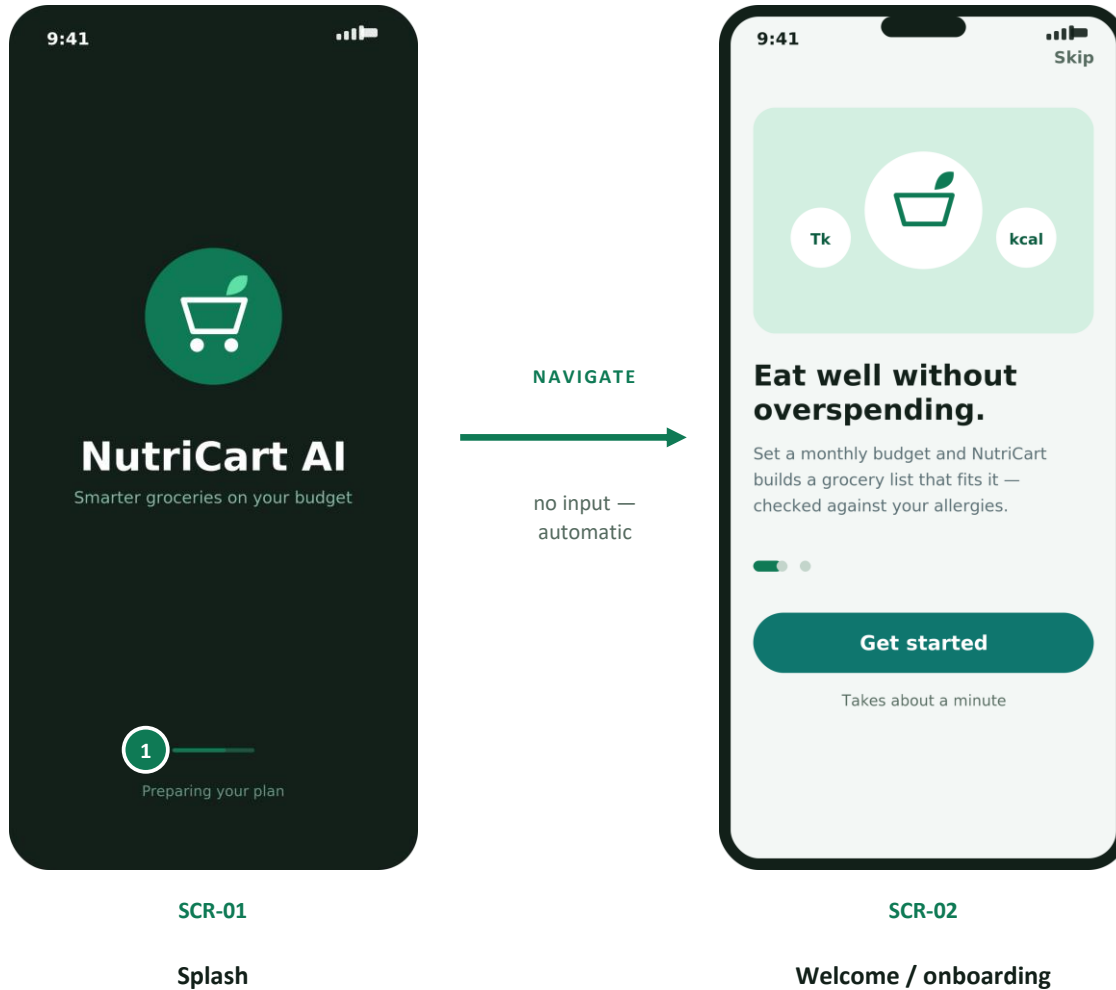
Flow numbers match the Complete Screen Flow Specification one-for-one, so FLOW-13 here is Flow 13 there.

Twelve screens and every forward path. Home is the hub; the generated list is the working surface everything else hangs off.



Teal = the AI generation step. ● = bottom-navigation root, reachable from any tab (FLOW-18 to FLOW-21). System back returns to the parent on every path and is specified per flow.

Application launch. The only branch in the app is decided before a screen is drawn.

**TRIGGER**

None. A LaunchedEffect resolves once local storage is open.

WHAT HAPPENS

ProfileDao.exists() is read. No profile row sends the user to onboarding; an existing profile goes straight to Home.

ROUTE

```
navigate("welcome") {  
  popUpTo("splash") { inclusive = true }  
}
```

ARGUMENTS

None

TRANSITION

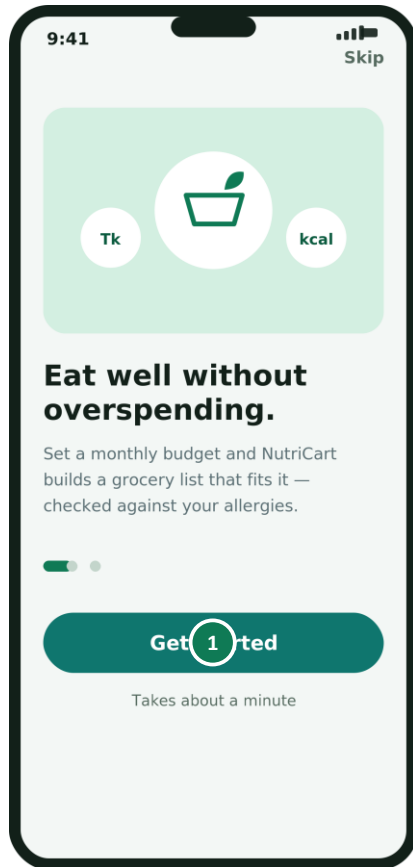
Fade, 200ms. No slide — the splash should not feel like a screen you left.

BACK BEHAVIOUR

Splash is removed from the stack. Back from Welcome exits the app.

The other branch uses the identical popUpTo with route "home", so the splash is never returnable either way.

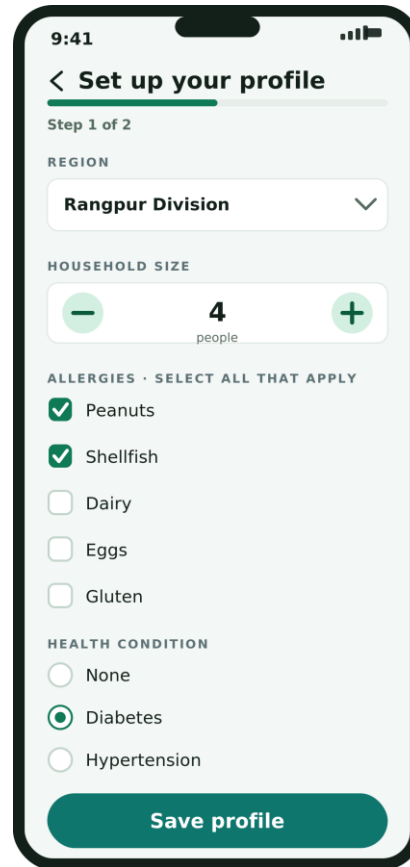
Start application. Onboarding hands straight over to profile setup.



SCR-02

Welcome / onboarding

NAVIGATE

tap
Get started

SCR-03

User profile setup

TRIGGER

Get started button (C-01) on the last onboarding pane.

WHAT HAPPENS

onboardingComplete is written, then the profile form opens empty.

ROUTE

```
navigate("profile/setup")
```

ARGUMENTS

None. A blank ProfileDraft is created by the ViewModel.

TRANSITION

Slide in from the right, 250ms.

BACK BEHAVIOUR

Pops to Welcome. Skip takes the same route but is recorded, so it is not offered again.

Complete user profile. Everything downstream depends on the four values saved here.

9:41

< Set up your profile

Step 1 of 2

REGION

Rangpur Division

HOUSEHOLD SIZE

4 people

ALLERGIES · SELECT ALL THAT APPLY

☒ Peanuts

☒ Shellfish

☐ Dairy

☐ Eggs

☐ Gluten

HEALTH CONDITION

☐ None

☒ Diabetes

☐ Hypertension

Save 1 profile

SCR-03

User profile setup

NAVIGATE

tap
Save profile

9:41

Good evening

Arpita

Seasonal picks

Cheaper this week in your region

MONTHLY BUDGET

Tk 12,000

Generate grocery list

Last month

Tk 11,240 spent · saved Tk 760

This month so far

42 ITEMS

78 NUTRI SCORE

2 ALERTS

Home List Nutrition Profile

SCR-04

Home dashboard

TRIGGER

Save profile button (C-01), enabled once region and health condition are both set.

WHAT HAPPENS

ProfileDao.upsert writes one row. Household size and region become inputs to every later generation.

ROUTE

```
navigate("home") {  
  popUpTo("welcome") { inclusive = true }  
}
```

ARGUMENTS

None. The profile is read from Room, not passed.

TRANSITION

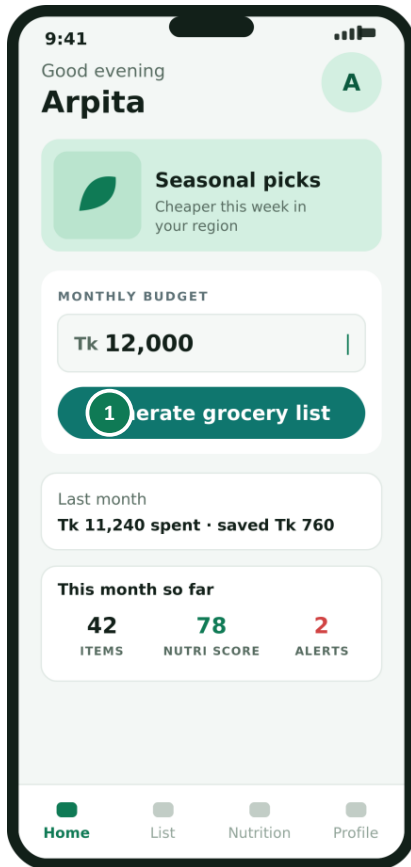
Slide in from the right, 250ms.

BACK BEHAVIOUR

Onboarding and setup are both cleared. Back on Home exits the app.

Reached a second time from FLOW-16, this screen returns to Profile instead — see FLOW-17. The route carries a mode argument to tell them apart.

Generate grocery list. One number is the entire input for a cycle.

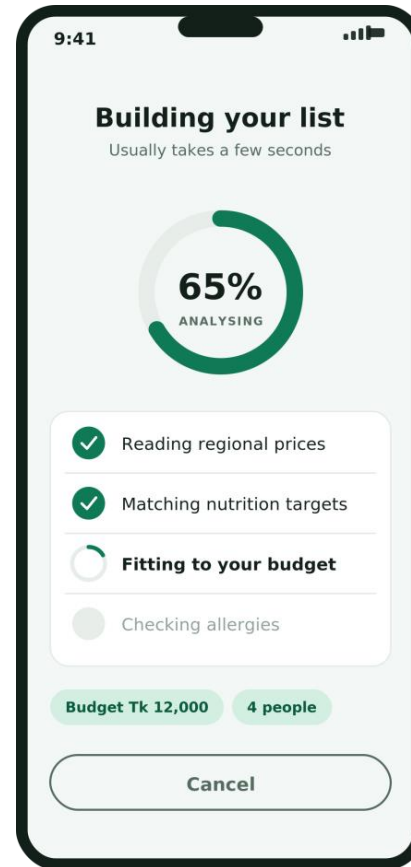


SCR-04

Home dashboard

NAVIGATE

enter budget,
tap Generate



SCR-10

AI processing

TRIGGER

Generate grocery list button (C-01), disabled while the budget field is blank.

WHAT HAPPENS

The budget is validated, then the generator starts. Processing is a real destination, not an overlay, so it survives rotation.

ROUTE

```
navigate("generate?budget={amount}")
```

ARGUMENTS

budget (Int, in Tk). Region, household size, allergies and condition are read from the profile.

TRANSITION

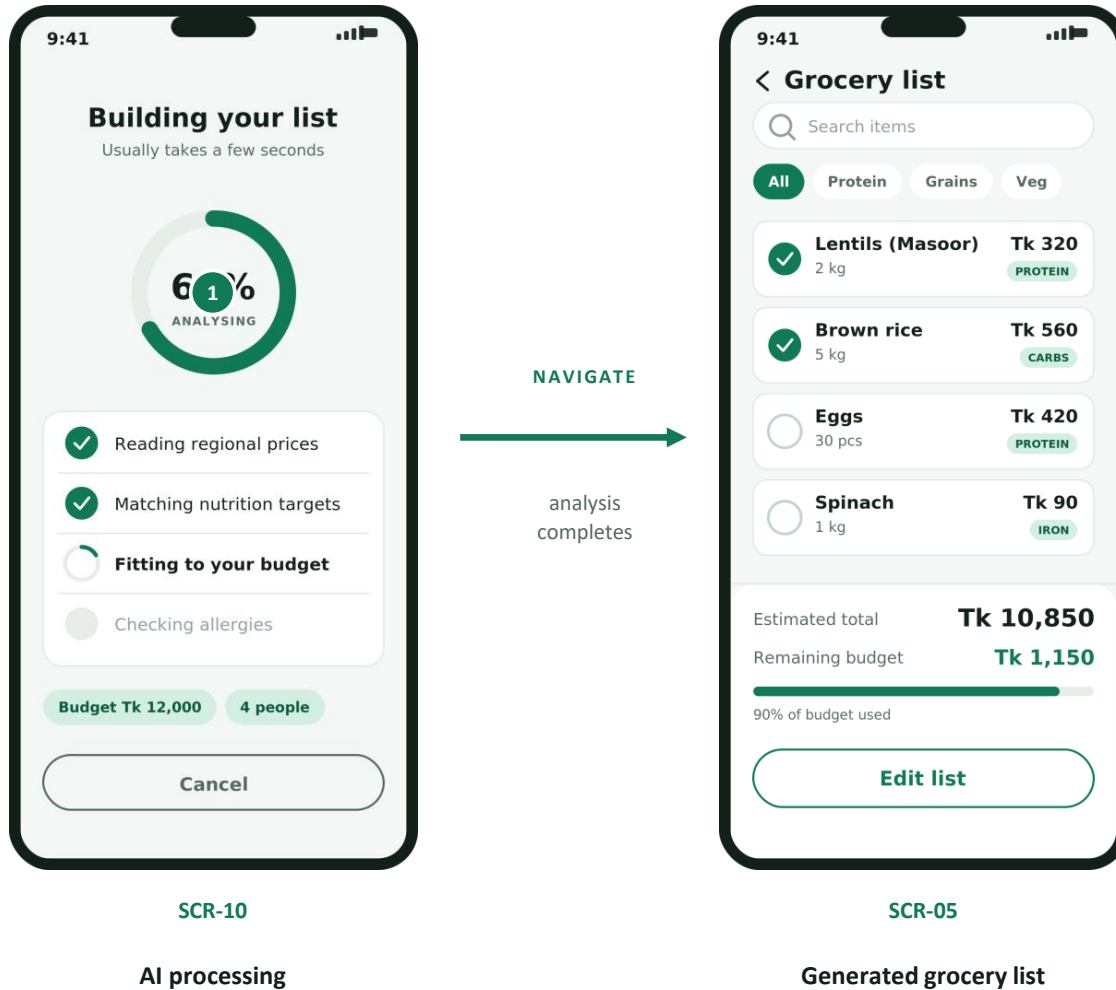
Fade through, 200ms — no slide, because the user has not moved anywhere yet.

BACK BEHAVIOUR

Back cancels generation and pops to Home. No partial list is written.

Making processing a route rather than a dialog is what lets a long generation survive a rotation or a backgrounded app.

AI grocery generation. Price, nutrition and budget analysis resolve into one list.

**TRIGGER**

None. The generator emits a completed list.

WHAT HAPPENS

Regional prices, nutrition targets for the household size, and the budget constraint are solved together, then written as one list with its items.

ROUTE

```
navigate("list/{listId}") {  
  popUpTo("generate") { inclusive = true }  
}
```

ARGUMENTS

listId (Long), returned by the insert.

TRANSITION

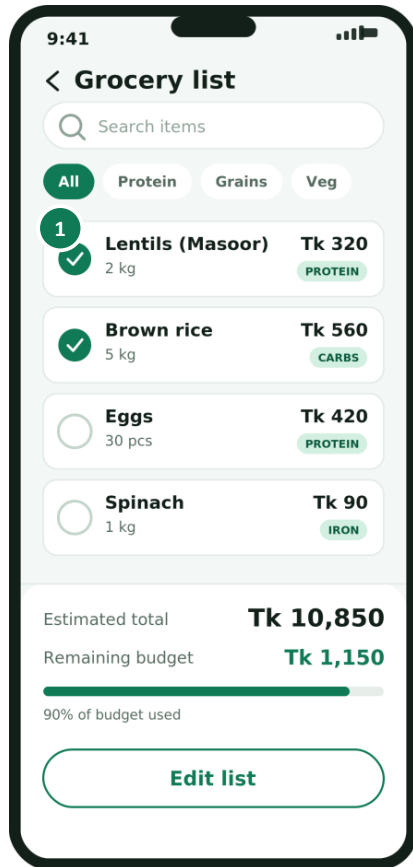
Crossfade, 250ms. Item rows stagger in at 40ms intervals.

BACK BEHAVIOUR

Processing is removed from the stack, so back from the list returns to Home.

On failure the screen stays put and shows a retry with the reason — an unaffordable balanced list is a result, not an error.

View grocery item details. Why an item was chosen, and what it contributes.

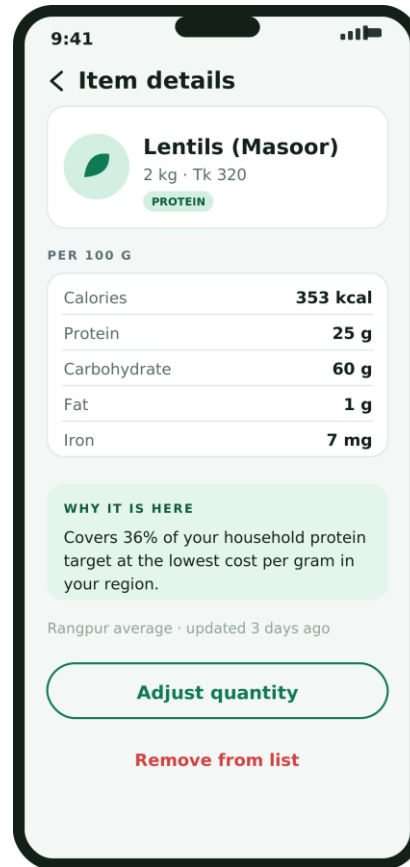


SCR-05

Generated grocery list

NAVIGATE

tap a
grocery item



SCR-11

Grocery item details

TRIGGER

Grocery item card (C-03). The checkbox inside it is a separate target and does not navigate.

WHAT HAPPENS

Opens the item's nutrition breakdown, its regional price basis, and its contribution to the household targets.

ROUTE

```
navigate("item/{itemId}")
```

ARGUMENTS

itemId (Long). Nutrition and price are re-queried, not passed.

TRANSITION

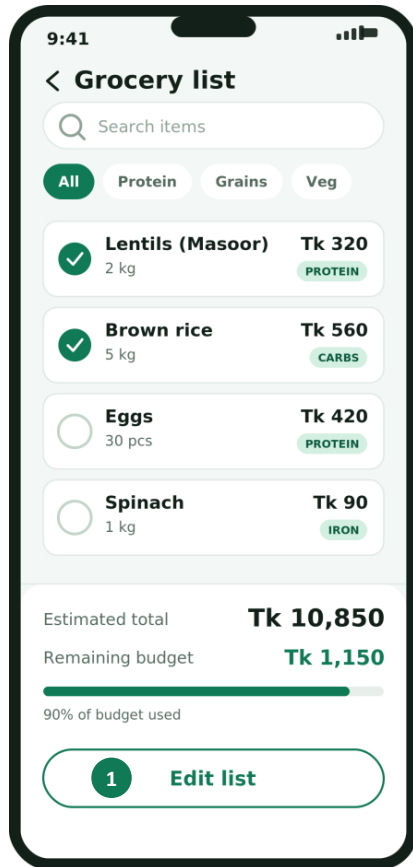
Shared-element on the item name, 300ms; container transform on the card.

BACK BEHAVIOUR

Pops one entry to the list.

Ticking the checkbox marks the item bought and dims the row without leaving the screen.

Edit grocery list. Manual correction of a generated result.

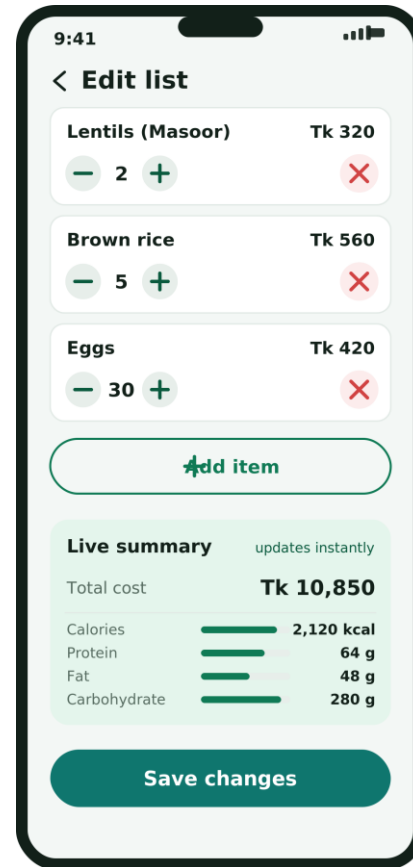


SCR-05

Generated grocery list

NAVIGATE

tap
Edit list



SCR-06

Edit grocery list

TRIGGER

Edit list button (C-02) in the docked total card.

WHAT HAPPENS

Opens the editable view. Quantities become steppers and each row gains a delete action.

ROUTE

```
navigate("list/{listId}/edit")
```

ARGUMENTS

listId (Long).

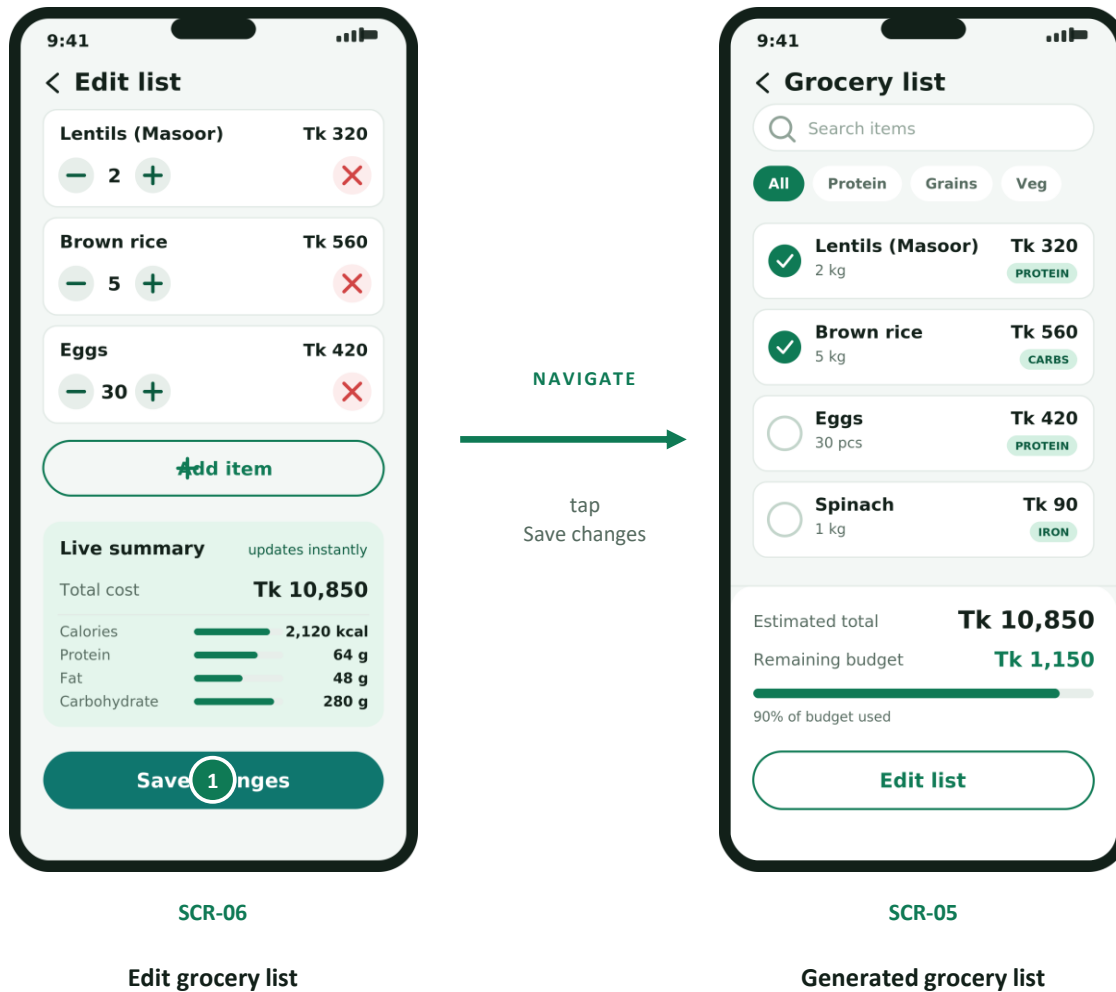
TRANSITION

Slide in from the right, 250ms.

BACK BEHAVIOUR

Warns before discarding unsaved changes, then pops to the list.

Save edited list. The updated list is the same screen with new numbers.



TRIGGER

Save changes button (C-01), docked above the inset.

WHAT HAPPENS

All quantity, deletion and addition changes are written in one transaction. Cost and nutrition were already recalculated live.

ROUTE

`popBackStack()`

ARGUMENTS

None. The list screen observes a Flow and updates itself.

TRANSITION

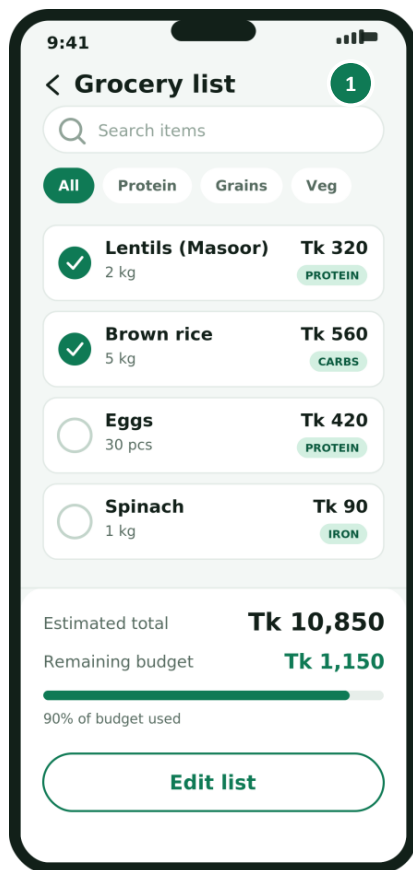
Slide out to the right, 250ms. Changed rows flash once over 300ms.

BACK BEHAVIOUR

Already handled — this is the back path.

The spec's "Updated Grocery List UI" is SCR-05 with new data, not a separate screen. Building it as one screen keeps a single source of truth for the totals.

Nutrition analysis. Scores the current list, generated or edited.

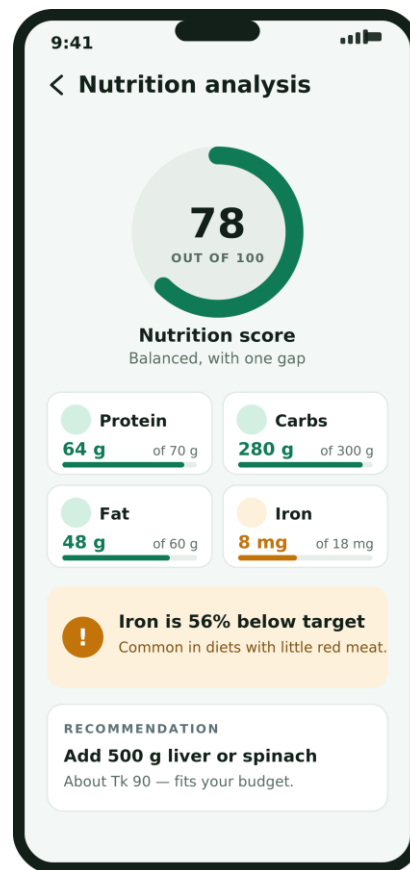


SCR-05

Generated grocery list

NAVIGATE

tap the
nutrition action



SCR-07

Nutrition analysis

TRIGGER

Nutrition action in the app bar of the list screen.

WHAT HAPPENS

NutritionCalculator scores the list as it stands right now against the RDA set for the household size.

ROUTE

```
navigate("nutrition/{listId}")
```

ARGUMENTS

listId (Long).

TRANSITION

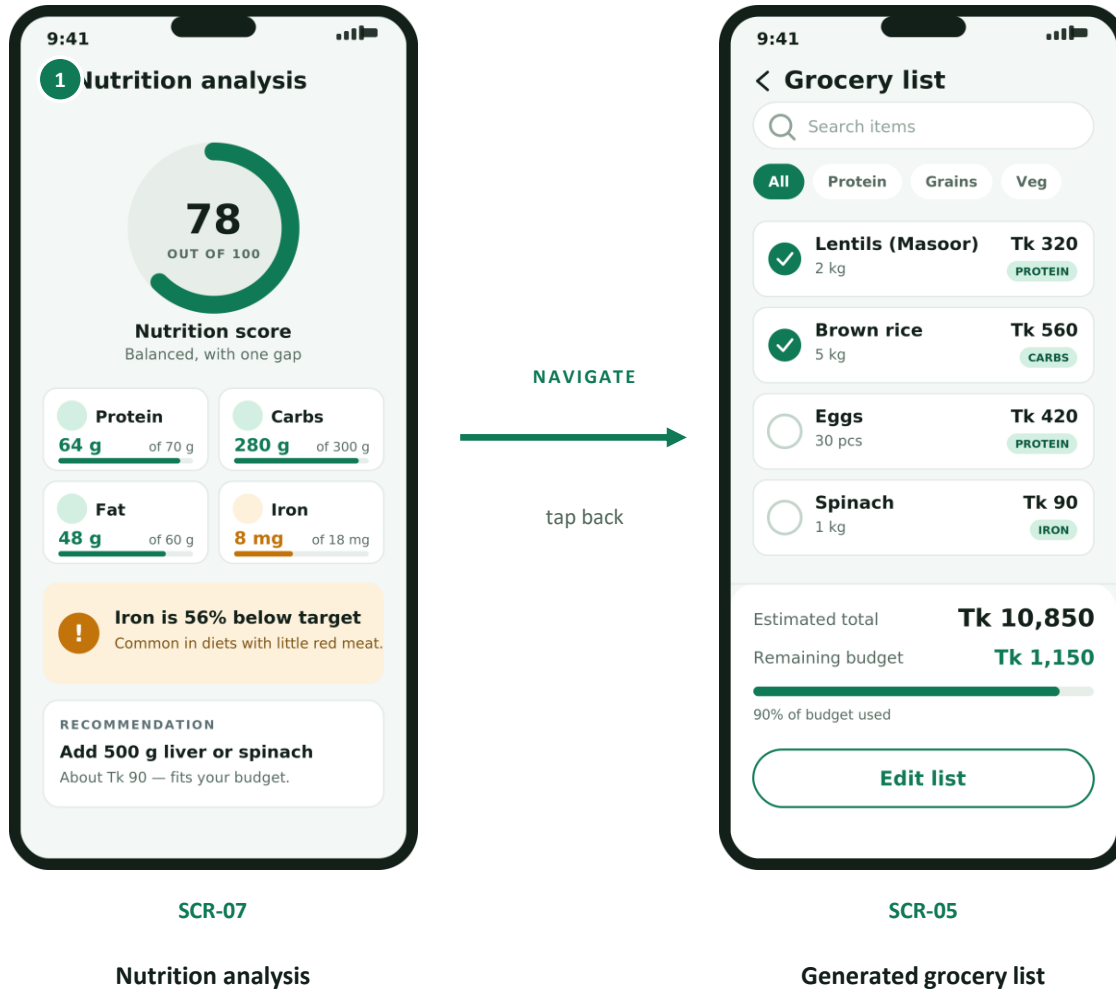
Slide up, 300ms. The score ring sweeps from zero over 400ms on arrival.

BACK BEHAVIOUR

Pops to the list — see FLOW-10.

This app-bar action is not yet drawn on the SCR-05 mockup. Add a 24dp chart icon as the trailing app-bar item when you update the Screen Details deck.

Return from nutrition analysis. Specified because the list may have changed underneath.



TRIGGER

Back chevron in the app bar, or the system back gesture.

WHAT HAPPENS

Pops to the list. Because the list is observed as a Flow, any edit made through a recommendation is already reflected.

ROUTE

`popBackStack()`

ARGUMENTS

None.

TRANSITION

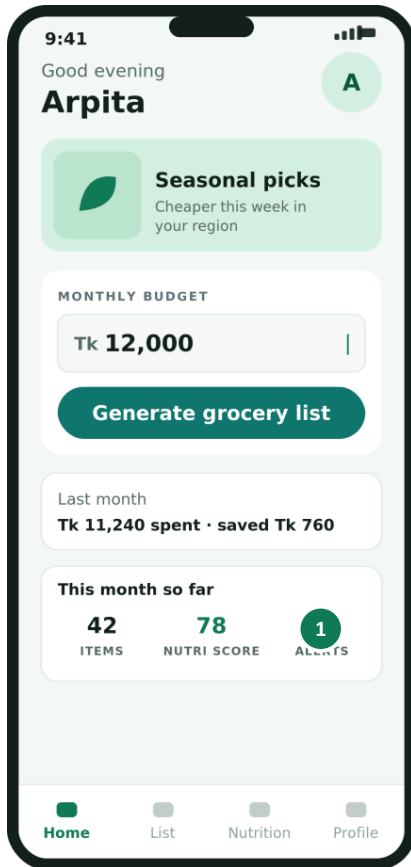
Slide down, 250ms — the reverse of FLOW-09.

BACK BEHAVIOUR

This is the back path. From the list, back continues to Home.

Nutrition is also a navigation root (FLOW-19). When entered that way there is no back chevron and the bottom nav stays visible.

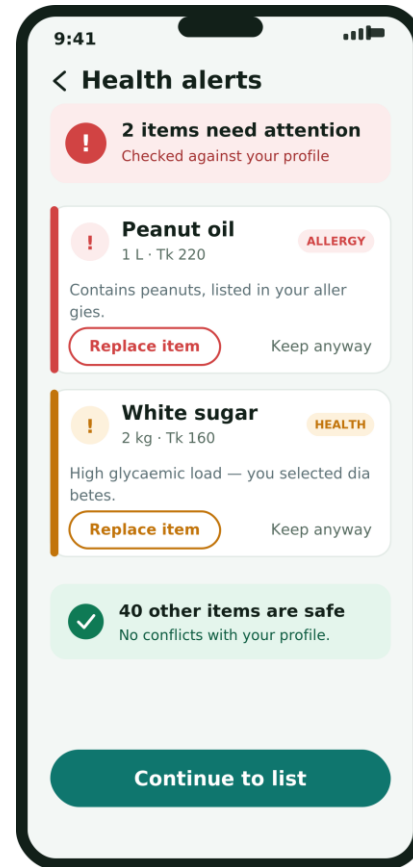
Allergy and health alerts. Reached from the count on the dashboard.



SCR-04

Home dashboard

NAVIGATE

tap the
alerts cell

SCR-08

Allergy & health alerts

TRIGGER

Alerts cell in the quick summary card, shown in the danger token when the count is above zero.

WHAT HAPPENS

AllergyMatcher runs over the current list and produces one card per conflicting item, allergies before health conditions.

ROUTE

```
navigate("alerts/{listId}")
```

ARGUMENTS

listId (Long). Allergies and condition come from the profile.

TRANSITION

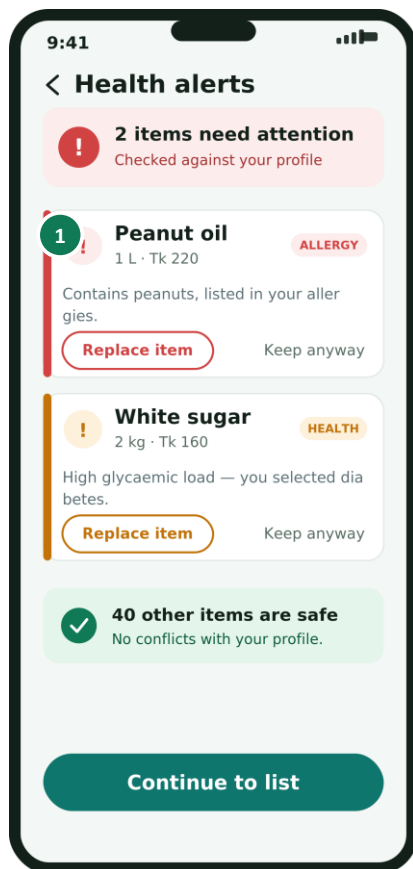
Slide in from the right, 250ms.

BACK BEHAVIOUR

Pops to Home.

The cell is not tappable when the count is zero — there is nothing to show, and the positive state already lives on SCR-08.

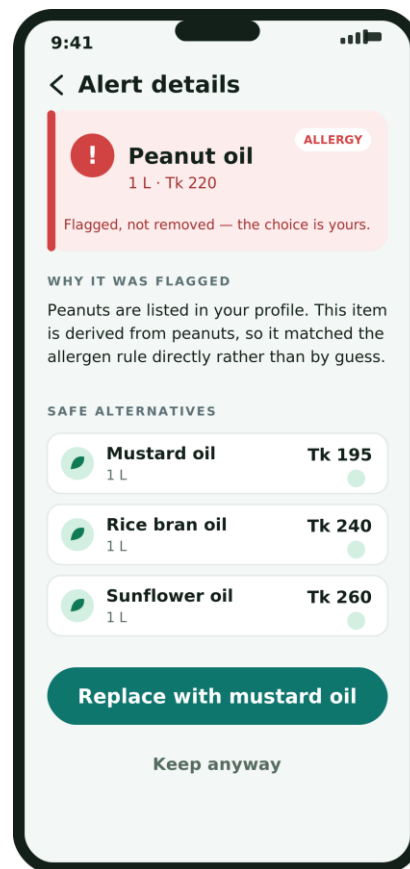
View alert details. The matched rule, and what to use instead.



SCR-08

Allergy & health alerts

NAVIGATE

tap a
warning item

SCR-12

Alert details

TRIGGER

Warning card (C-03) on the alerts screen.

WHAT HAPPENS

Opens the full explanation: which profile value matched, why, and a priced list of safe alternatives.

ROUTE

```
navigate("alerts/{listId}/item/{itemId}")
```

ARGUMENTS

listId and itemId (both Long).

TRANSITION

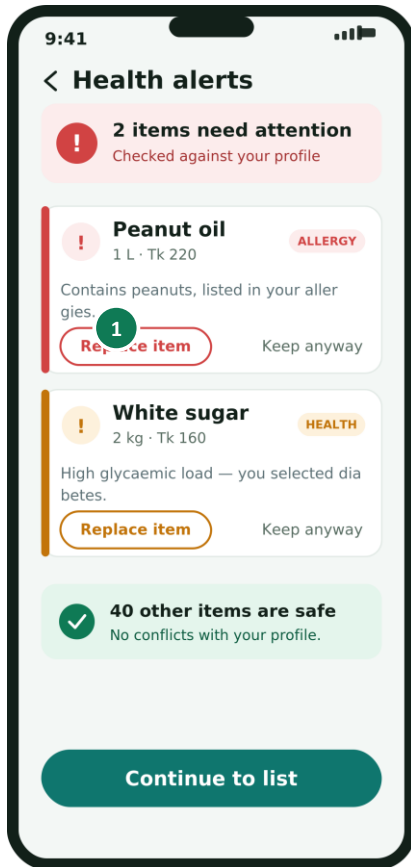
Slide in from the right, 250ms.

BACK BEHAVIOUR

Pops to the alerts list.

Alternatives are ranked by price proximity, so a replacement does not quietly break the budget.

Replace unsafe item. The list updates and the alert is resolved.

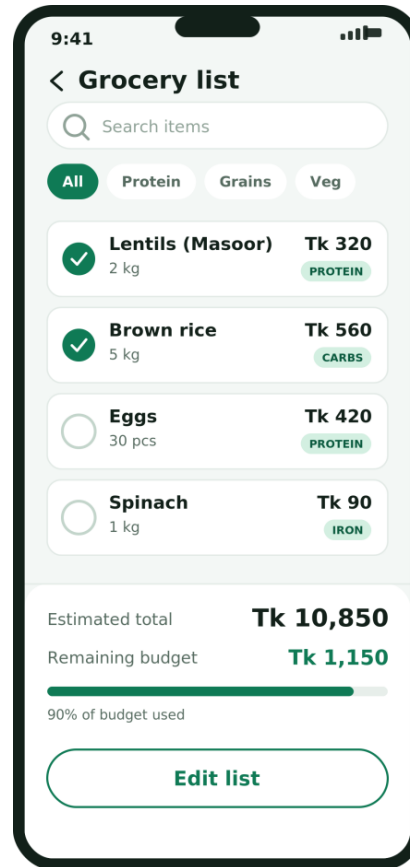


SCR-08

Allergy & health alerts

NAVIGATE

tap
Replace item



SCR-05

Generated grocery list

TRIGGER

Replace item button (C-02) on a warning card, or the primary button on SCR-12.

WHAT HAPPENS

The flagged item is swapped for the chosen alternative. Cost, nutrition and the alert count all recompute.

ROUTE

`popBackStack()` with a replacement result

ARGUMENTS

itemId replaced and replacementId chosen, returned through a `SavedStateHandle`.

TRANSITION

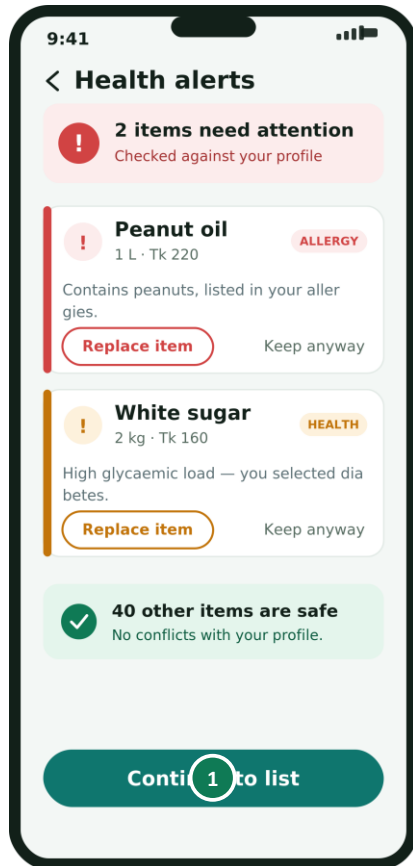
The card collapses over 250ms, then the screen pops if no alerts remain.

BACK BEHAVIOUR

If other alerts remain, the user stays on SCR-08 rather than being ejected.

The spec's "Grocery List Update UI" is SCR-05 after the write. Same screen, new totals.

Continue with warning. Flagged, not removed — the choice stays with the user.

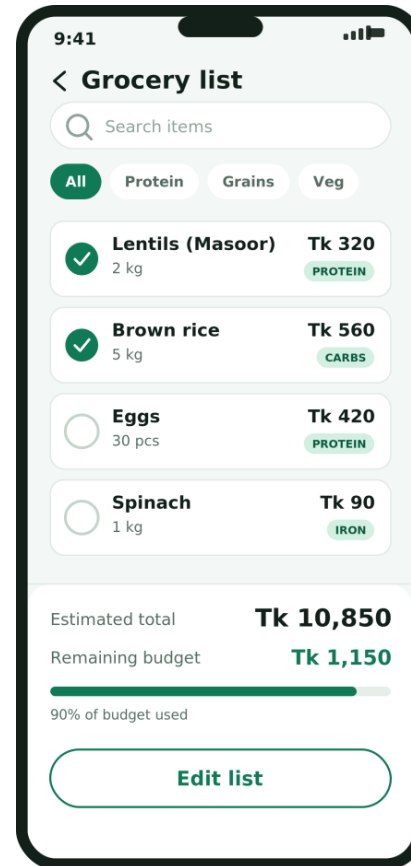


SCR-08

Allergy & health alerts

NAVIGATE

tap
Continue



SCR-05

Generated grocery list

TRIGGER

Continue to list button (C-01), or Keep anyway on an individual card.

WHAT HAPPENS

An override is recorded against each acknowledged item so the same alert does not reappear for this list.

ROUTE

```
popBackStack()
```

ARGUMENTS

None. Overrides are written before the pop.

TRANSITION

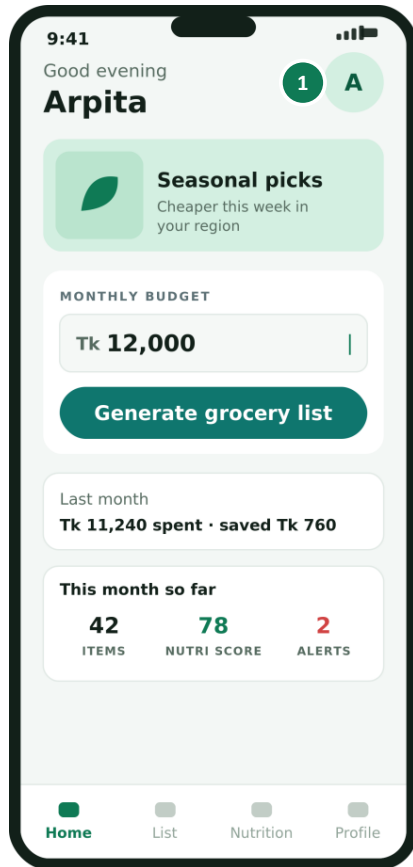
Slide out to the right, 250ms.

BACK BEHAVIOUR

Returns to the list with every item intact, flagged ones included.

This is the abstract's design position made concrete: conflicts are surfaced and explained, never silently deleted.

Profile access from the dashboard avatar.

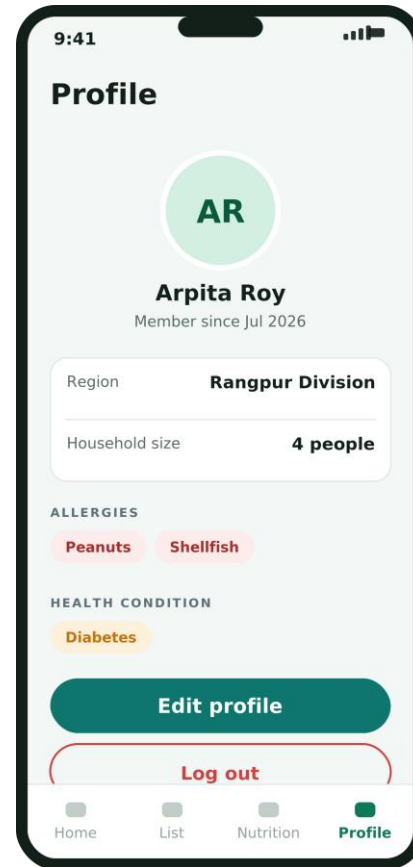


SCR-04

Home dashboard

NAVIGATE

tap the
avatar



SCR-09

Profile

TRIGGER

User avatar in the dashboard header, 88dp with a 48dp minimum target.

WHAT HAPPENS

Opens the profile root showing region, household size, allergies and condition as read from the single profile row.

ROUTE

```
navigate("profile") {  
  popUpTo(startDestination) {  
    saveState = true }  
  launchSingleTop = true  
  NonstoreState = true  
}
```

TRANSITION

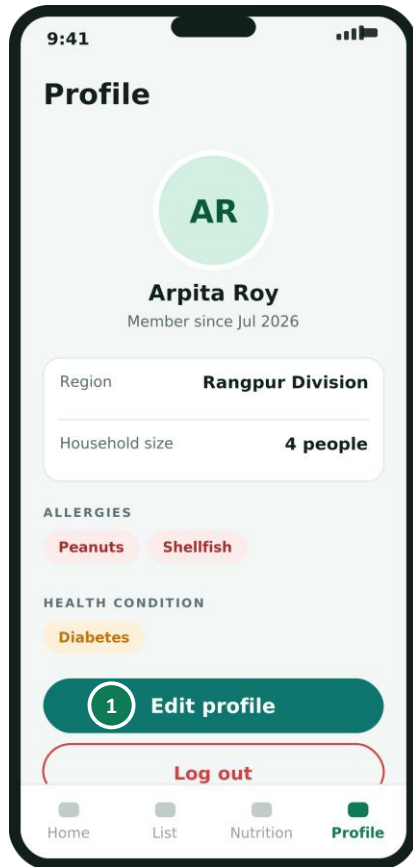
Fade, 150ms — Profile is a nav root, so it does not slide.

BACK BEHAVIOUR

Returns to Home.

The avatar and the Profile tab (FLOW-20) resolve to the same destination, so the tab stays correctly selected either way.

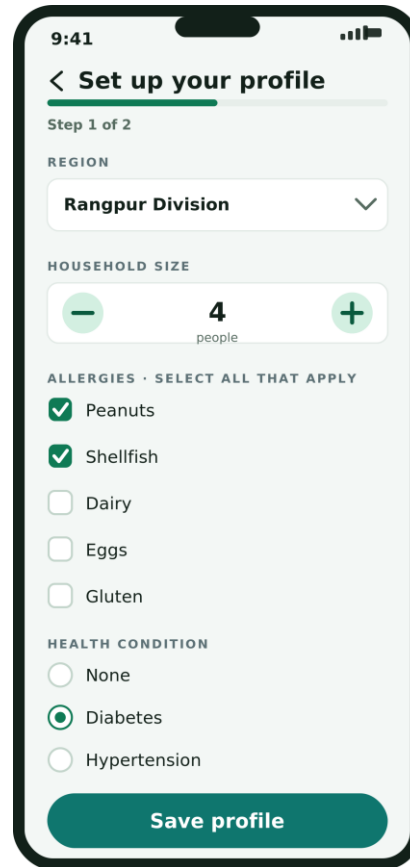
Edit profile. The setup form reopened with values loaded.



SCR-09

Profile

NAVIGATE

tap
Edit profile

SCR-03

User profile setup

TRIGGER

Edit profile button (C-01).

WHAT HAPPENS

The setup form opens pre-filled in edit mode. The step indicator is hidden because there is no second step when editing.

ROUTE

```
navigate(  
  "profile/setup?mode=edit"  
)
```

ARGUMENTS

mode = edit. The profile row is read inside the ViewModel.

TRANSITION

Slide in from the right, 250ms.

BACK BEHAVIOUR

Warns before discarding changes, then pops to Profile.

Changing region or household size invalidates the current list's pricing. The save path warns and offers to regenerate.

Update profile. Same screen as FLOW-03, different exit.

9:41

< Set up your profile

Step 1 of 2

REGION

Rangpur Division

HOUSEHOLD SIZE

4 people

ALLERGIES · SELECT ALL THAT APPLY

☒ Peanuts

☒ Shellfish

☐ Dairy

☐ Eggs

☐ Gluten

HEALTH CONDITION

☐ None

☒ Diabetes

☐ Hypertension

Save 1 profile

SCR-03

User profile setup

NAVIGATE

modify, then
tap Save

9:41

Profile

AR

Arpita Roy

Member since Jul 2026

Region Rangpur Division

Household size 4 people

ALLERGIES

Peanuts Shellfish

HEALTH CONDITION

Diabetes

Edit profile

Log out

Home List Nutrition Profile

SCR-09

Profile

TRIGGER

Save profile button (C-01).

WHAT HAPPENS

ProfileDao.upsert overwrites the single row. Profile is observing it, so the new values appear without a refresh.

ROUTE

popBackStack()

ARGUMENTS

None.

TRANSITION

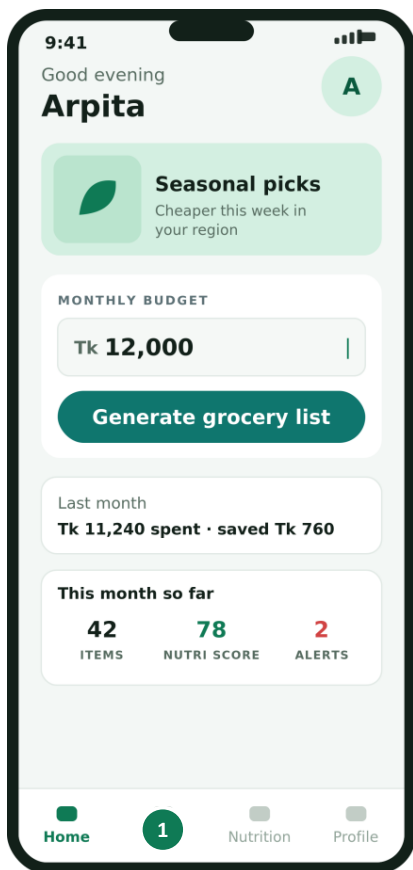
Slide out to the right, 250ms.

BACK BEHAVIOUR

Already handled — this is the back path.

FLOW-03 navigates forward to Home, FLOW-17 pops back to Profile. The mode argument set in FLOW-16 is what decides which.

Grocery list navigation. Tab switching between four roots.

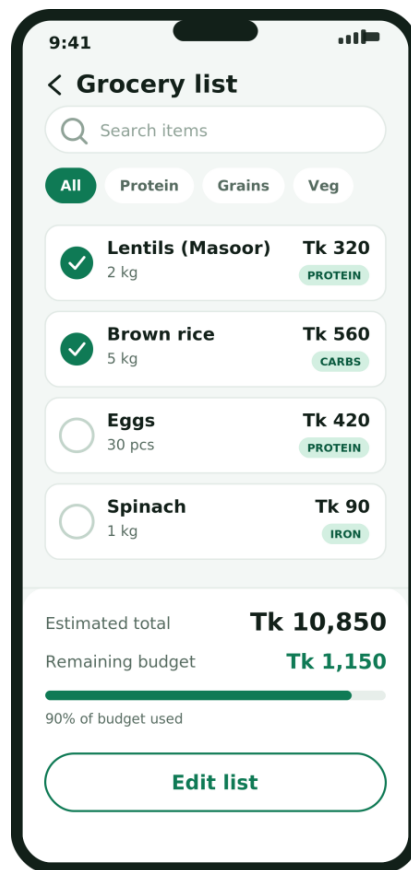


SCR-04

Home dashboard

NAVIGATE

tap the
List tab



SCR-05

Generated grocery list

TRIGGER

Bottom navigation (C-12), second item.

WHAT HAPPENS

Switches the nav root to the most recent list. Each tab keeps its own scroll position and state.

ROUTE

```
navigate("list") {  
  popUpTo(startDestination) {  
    saveState = true }  
  }  
  launchSingleTop = true  
  restoreState = true  
}
```

TRANSITION

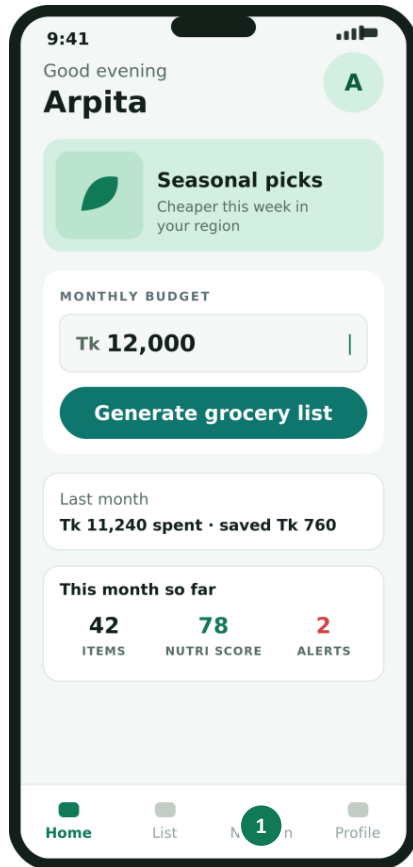
Fade, 150ms. Never a slide — tabs are siblings, not steps.

BACK BEHAVIOUR

Returns to Home from any tab. Tabs never stack on each other.

SCR-05 is reachable two ways: pushed from FLOW-05 with a back chevron, or as a tab root with the bottom nav visible. Same composable, two app-bar configurations.

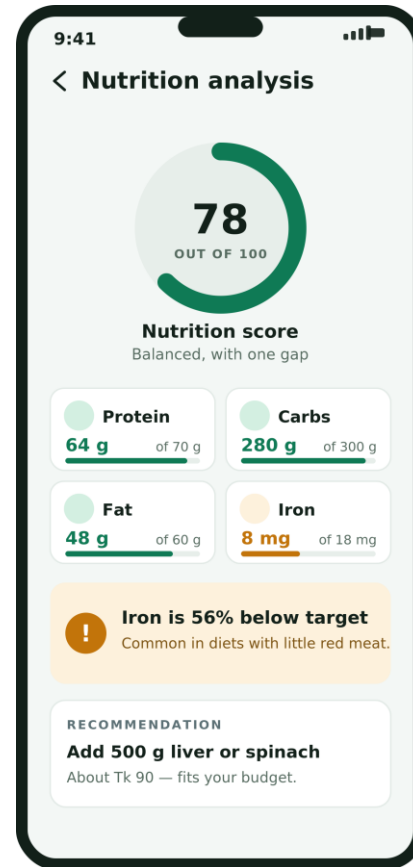
Nutrition navigation. The same screen FLOW-09 reaches, entered as a root.



SCR-04

Home dashboard

NAVIGATE

tap the
Nutrition tab

SCR-07

Nutrition analysis

TRIGGER

Bottom navigation (C-12), third item.

WHAT HAPPENS

Scores the most recent list. Identical content to FLOW-09; only the entry point differs.

ROUTE

```
navigate("nutrition") {  
  popUpTo(startDestination) {  
    saveState = true }  
  launchSingleTop = true  
  NonstoreState = true  
}
```

TRANSITION

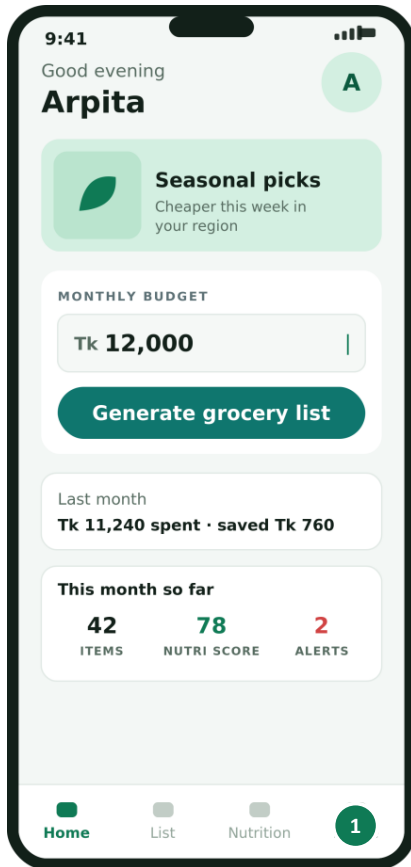
Fade, 150ms.

BACK BEHAVIOUR

Returns to Home.

Disabled with a one-line explanation when no list has been generated yet — there is nothing to score.

Profile navigation. The fourth root.

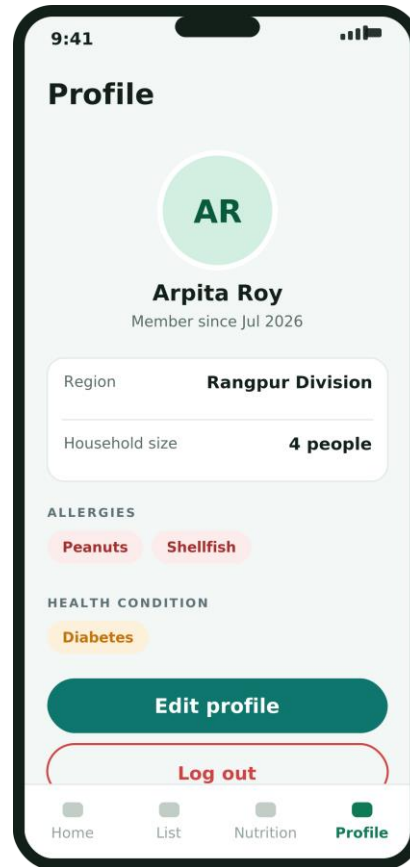


SCR-04

Home dashboard

NAVIGATE

tap the
Profile tab



SCR-09

Profile

TRIGGER

Bottom navigation (C-12), fourth item.

WHAT HAPPENS

Opens the profile root. Preference writes on that screen are immediate — there is no save step.

ROUTE

```
navigate("profile") {  
  popUpTo(startDestination) {  
    saveState = true }  
  launchSingleTop = true  
  NonstoreState = true  
}
```

TRANSITION

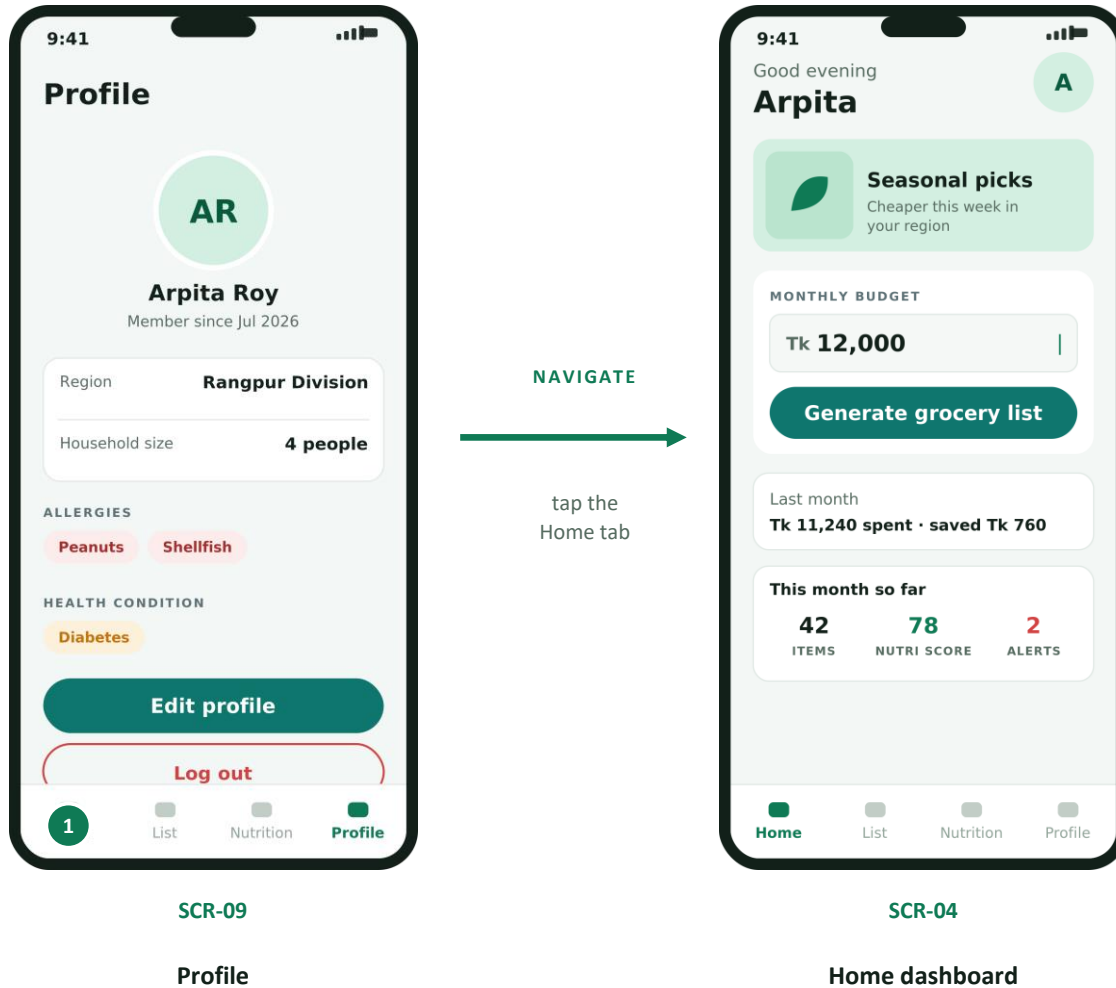
Fade, 150ms.

BACK BEHAVIOUR

Returns to Home.

Identical destination to FLOW-15. Specify the nav block once and reuse it for all four tabs.

Home navigation. The start destination, reachable from every tab.



TRIGGER

Bottom navigation (C-12), first item.

WHAT HAPPENS

Returns to the dashboard with its budget field and summary restored from saved state.

ROUTE

```
navigate("home") {  
  popUpTo(startDestination) {  
    saveState = true  
  }  
  launchSingleTop = true  
  restoreState = true  
}
```

TRANSITION

Fade, 150ms.

BACK BEHAVIOUR

Home is the start destination — back exits the app.

restoreState = true is what makes a typed but unsubmitted budget survive a trip through another tab.